Mathematical Statistics II

Point Estimation: Principles and Theory

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Introduction

Introduction

- We now introduce principles of parameter estimation, as well as associated theory.
- The three approaches we have considered thus far (MoM, MLE, Bayes) are great starting points, but we would like to study general properties of parameter estimations, and how these approaches align with these properties.
- There is no "correct" order in which these material are presented, so we deviate a bit from our textbooks.
- This work is largely based on Rice (2007, Sections 8.5.2, 8.7–8.8), Casella and Berger (2024, Chapters 6, 7, 10), as well as some supplemental material from Pawitan (2001).

Fisher's Information

Curvature of the Likelihood Function

- Both the MLE and Bayesian estimates are forms of Likelihood based inference.
- For the MLE, we are interested in the maximum of the likelihood function.
- For Bayesian inference, the posterior distribution is the (scaled) product of the likelihood and a prior distribution:

$$\pi(\theta|x) \propto L(\theta) \times \pi(\theta).$$

- In both instances, the shape of the likelihood function $L(\theta)$ is of interest.

Curvature of the Likelihood Function II

- Coming up: We'll introduce the *score* function, and *Fisher's Information*. The latter can be used to obtain uncertainty estimates for *any* parameter estimated using the MLE technique.

Curvature of the Likelihood Function III

Definition: The score function

Let $L(\theta) = f(x^*; \theta)$ be the likelihood of some model, conditioned on the observed data (x^* is assumed to be multi-variate). Then we define the score function $S(\theta)$ to be:

$$S(\theta) = \frac{\partial}{\partial \theta} \log L(\theta).$$

- Assuming the MLE $\hat{\theta}$ exists and is not a boundary point, then:

$$S\left(\hat{\theta}\right) = 0.$$

- As written, the score function is dependent on a fixed data set: x^*.

Curvature of the Likelihood Function IV

- We could also expand this by considering it as a random function of any possible data X, which is random.

Curvature of the Likelihood Function V

- Since we are interested in the curvature of the likelihood function, we should also consider the second derivative.
- At the MLE, the second derivative of the log-likelihood will be *negative*, so it's natural to consider the negative second derivative of the log-likelihood:

$$I(\theta) = -\frac{\partial^2}{\partial \theta^2} \log L(\theta).$$

- Again, we are treating the likelihood function $L(\theta)$ as a regular (non-random) function, fixed on x^* (this is slightly different than many texts).
- In this case, we call $I(\theta)$ the observed Fisher's Information, at the point θ.

Curvature of the Likelihood Function VI

Comment: joint information

We can define the information for just a single point $X_i = x_i^*$,

$$I_i(\theta) = -\frac{\partial^2}{\partial\theta^2} \log f_{X_i}(x_i^*; \theta),$$

or for the entire collection: $X_{1:n} = x_{1:n}^*$:

$$I(\theta) = -\frac{\partial^2}{\partial\theta^2} \sum_i \log f_{X_i}(x_i^*; \theta) = \sum_i I_i(\theta).$$

- The curvature at the MLE is of particular interest.

Curvature of the Likelihood Function VII

Definition: The Observed Fisher Information

Let $L(\theta)$ be the likelihood function, fixed on the observed data x^*. Unless otherwise specified, the observed Fisher information is understood to be the negative second derivative of $\log L(\theta)$, evaluated at the MLE:

$$I\left(\hat{\theta}\right) = -\frac{\partial^2}{\partial\theta^2} \log L\left(\theta\right)\Big|_{\theta=\hat{\theta}}$$

Note that when θ is multi-variate, this is a matrix with entries:

$$\left[I\left(\hat{\theta}\right)\right]_{i,j} = -\frac{\partial^2}{\partial\theta_i \partial\theta_j} \log L(\theta)\Big|_{\theta=\hat{\theta}},$$

also denoted $I\left(\hat{\theta}\right) = -\nabla^2 \log L\left(\hat{\theta}\right)$.

Curvature of the Likelihood Function VIII

- Note that large values of $I\left(\hat{\theta}\right)$ in the univariate case (and large eigen-values / determinant in the multivariate case) indicate a "tight peak" around the maximum, suggesting less uncertainty about the parameter θ.

Observed Fisher's information, Binomial Distribution

Let $X \sim \text{Binom}(n, \theta)$. Find the MLE by setting the score function equal to zero, and then use that to compute the observed Fisher's information.

Curvature of the Likelihood Function IX

Observed Fisher's information, Normal Distribution

Suppose $X_1, \ldots, X_n$ are iid from $N(\theta, \sigma^2)$, with known σ^2. Find the MLE by setting the score function to zero, and then calculate the joint information, $I_n(\theta)$ at the MLE.

Curvature of the Likelihood Function X

- Calculating the observed Fisher's Information at any given point is rather simple: Find the likelihood, take second derivative(s), plug in desired point.
- However, because data are random, we may want to consider the expected Fisher's information.

Curvature of the Likelihood Function XI

Theorem: Expected Score

Now consider treating the score function $S(\theta)$ as a random function, dependent on random input data X. That is:

$$S(\theta) = \frac{\partial}{\partial\theta} \log f(X;\theta).$$

Then, under mild regularity conditions, the expected score is zero:

$$E\big[S(\theta)\big] = 0.$$

- NOTE: This is the most common way to define the score (and later, information). This holds for either a single observation, or the entire collection of data.

Curvature of the Likelihood Function XII

Definition: Fisher's Information

Continuing with the treatment of the score function as a random function of X, we will define the Expected Fisher's Information to be the following:

$$\mathcal{I}(\theta) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big] = E\Big[\big(S(\theta)\big)^2\Big].$$

This is often referred to as just Fisher's Information.

Curvature of the Likelihood Function XIII

Proposition: Variance of the Score

If the partial derivative and integral operations can be exchanged, then

$$\mathcal{I}(\theta) = \mathrm{Var}\big(S(\theta)\big) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big],$$

where the score function is treated as random.

Curvature of the Likelihood Function XIV

- This proposition is useful because:
  - If the log-likelihood is differentiable, the first definition $\mathcal{I}(\theta) = E[S(\theta)^2]$ always exists.
  - Typically, we can express this as $\mathcal{I}(\theta) = \mathrm{Var}(S(\theta))$, which is a nice interpretable quantity.
  - The final expression $\mathcal{I}(\theta) = -E\left[\frac{\partial^2}{\partial\theta^2} \log f(X;\theta)\right]$ is usually the easiest to calculate.

Curvature of the Likelihood Function XV

- For the multivariate case, this is equivalent to writing:

$$
\begin{aligned}
\big[\mathcal{I}(\theta)\big]_{i,j} &:= E_\theta\Big[\frac{\partial \log f(X;\theta)}{\partial \theta_i}\frac{\partial \log f(X;\theta)}{\partial \theta_j}\Big] \\
&= \mathrm{Cov}\Big(\frac{\partial \log f(X;\theta)}{\partial \theta_i}, \frac{\partial \log f(X;\theta)}{\partial \theta_j}\Big) \\
&= -E\Big[\frac{\partial^2}{\partial \theta_i \partial \theta_j}\log f(X;\theta)\Big]
\end{aligned}
$$

Curvature of the Likelihood Function XVI

- Note that the curvature $I(\theta)$, defined as the second derivative of $L(\theta)$ (fixed function based on the observed data), exists for all points θ.
- Many theorem's instead rely on $\mathcal{I}(\theta)$, which is the expected Fisher's information. This later quantity "integrates out" the effect of the randomness in the data.
- In practice, the expected information is often hard to find, while the observed information is easy to calculate. For instance, in cases where the expected information is available, we might be able to get exact confidence intervals, which are preferred.
- In practice, we often replace $\mathcal{I}(\hat{\theta})$ in theory with $I(\hat{\theta})$.

Curvature of the Likelihood Function XVII

- However, we're primarily only interested in this function at the MLE, *or* when we regard $S(\theta)$ as a random function of arbitrary X (not fixed on x^*).

Curvature of the Likelihood Function XVIII

Comment on numeric optimization

If numeric optimization is used to find the MLE, many approaches estimate the Hessian of the loss-function. If the negative log-likelihood is used, setting `optim(..., hessian = TRUE)` can be used to compute the Observed Fisher's Information (don't forget to adjust for variable transformations!)

Asymptotic proprties of the MLE and Bayesian estimators

Normality of MLE

- Under weak regularity conditions, both the MLE and Bayesian posterior converge to a Normal distribution, centered at the true value.
- We'll begin with a sketch proof of the distribution of the MLE.

Theorem: Asymptotic distribution of the MLE

Let $x_1^*, x_2^*, \ldots, x_n^*$ come from a family of distributions with density $f(x;\theta)$. If the true value of θ is θ_0, i.e., $X_i \overset{iid}{\sim} f(x;\theta_0)$, and we denote $\hat{\theta}$ as the MLE of θ, then

$$\sqrt{n\mathcal{I}(\theta_0)}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z$$

where $Z \sim N(0,1)$.

Normality of MLE II

- How do we interpret / use the result above?
- For large iid samples: the MLE $\hat{\theta}$ is approximately normally distributed, centered at the true value θ_0, with shrinking variance:
$$\mathrm{Var}(\hat{\theta}) = \frac{1}{n\mathcal{I}(\theta_0)}.$$
- Note that there is sometimes some confusion regarding whether or not $\mathcal{I}$ is the information from *a single* observation, or *all observations*.
- Because the data are IID, it's really just a difference in scaling constants.

Normality of MLE III

- In our definition and proof, we let $\mathcal{I}(\theta_0)$ be the information from a single observation:

$$\mathcal{I}(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X_i;\theta)\Big)^2\Big].$$

- We could have instead considered the likelihood / information using all of the data points:

$$\mathcal{I}_n(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f_{X_1:X_n}(X_1,\ldots,X_n;\theta)\Big)^2\Big].$$

Normality of MLE IV

- In the later case, the theorem would have stated:

$$\sqrt{\mathcal{I}_n}(\hat{\theta} - \theta_0) \xrightarrow{d} Z \sim N(0, 1),$$

  or that

$$\mathrm{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}_n(\theta_0)}$$

- In the IID case, these are the same, as

$$\begin{aligned}\mathcal{I}_n(\theta) &= -E\Big[\frac{\partial^2}{\partial\theta^2}\sum_i \log f(X_i;\theta)\Big] \\ &= -\sum_i E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta)\Big] = n\mathcal{I}(\theta).\end{aligned}$$

Normality of MLE V

- Our derivation treated θ as univariate. The more common multivariate case has a similar result.

Multivariate parameters

If $\theta \in \mathbb{R}^d$ is multivariate, than using the matrix definition of $\mathcal{I}(\theta)$, then:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z_d \sim N_d\big(0_d, \mathcal{I}(\theta_0)^{-1}\big),$$

where $\hat{\theta}$ is the MLE, θ_0 is the true parameter value, and $N_d(\mu, \Sigma)$ is a d-dimensional multivariate normal distribution with mean vector μ and covariance matrix Σ.

Implications for estimating parameter uncertainty

- This result gives us an immediate way to approximate the uncertainty associated with the MLE, for rather arbitrary models.
- If $\hat{\theta}$ is the MLE, then:

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{n\mathcal{I}(\hat{\theta})} = \frac{1}{\mathcal{I}_n(\hat{\theta})}.$$

- There are two different approximations here:
  - Asymptotic normality.
  - $\hat{\theta} \xrightarrow{p} \theta_0$ (Slutksy's Theorem).
- It is a good approximation for even moderate n, but exact calculations are preferred when possible.

Implications for estimating parameter uncertainty II

- Finally, recall that $\mathcal{I}(\hat{\theta})$ is an expectation.
- When an exact calculation is not readily available, calculating the expectation $\mathcal{I}(\hat{\theta})$ usually isn't either!
- In practice, we often replace the Fisher's Information with the Observed Fisher's information:

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{I_n(\hat{\theta})},$$

  where I_n is the information provided by all n data points.
- The details justifying this final approximation are a bit tricky. See Keener (2010, Chapter 9) for more details.
- In short, for many distributions (regular exponential families), $\mathcal{I}(\hat{\theta}) = I(\hat{\theta})$. In other cases, suitable conditions imply that $I(\hat{\theta}) \xrightarrow{p} \mathcal{I}(\hat{\theta})$.

Asymptotic results for Bayesian posteriors

- Similar results hold for Bayesian estimates.

Bernstein–von Mises Theorem

We are not ready to state the full Theorem, so we'll just provide a heuristic approach: under suitable conditions, the posterior distribution behaves asymptotically like a normal distribution with mean $\hat{\theta}_{MLE}$, and variance $1/\mathcal{I}(\theta_0)$.

Bias, MSE, and Consistency

Bias

- We have already talked a little about bias last semester.
- It's an important concept / principle for point-estimates.

Definition: Bias of an estimator

The bias of a point estimator $\hat{\theta}$ of a parameter θ is the difference between the expected value of $\hat{\theta}$ and θ; that is,

$$\text{Bias}_\theta\left[\hat{\theta}\right] = E_\theta\left[\hat{\theta}\right] - \theta$$

- If $\text{Bias}_\theta\left[\hat{\theta}\right] = 0$, then $\hat{\theta}$ is an unbiased estimator of θ, and $E_\theta\left[\hat{\theta}\right] = 0$.

Bias II

- unbiasedness is a nice property, but alone does not mean we have a good estimator.
- Another thing that needs to be considered is the variance of an estimator.
- We encountered this with Monte-Carlo methods: many approaches converge to the true value, but lower variance estimates are generally preferred among unbiased methods.

Bias III

Unbiased estimators for normal distribution

Let $X_1, X_2, \ldots, X_n$ be iid from $N(\mu, \sigma^2)$. Consider two estimates of the parameter $\theta = \mu$:

$$\hat{\theta}_1 = \bar{X}_n, \qquad \hat{\theta}_2 = X_1.$$

Show that both are unbiased estimators, but that $\hat{\theta}_1$ has lower variance.

MSE

- The mean square error (MSE) describes the expected squared difference between an estimator and the true parameter value.
- Often, attempts as lowering the *variance* results in increasing the *variance*.
- It can be used to balance the bias-variance tradeoff.

Definition + Theorem: MSE

Let $\hat{\theta}$ be an estimator of θ. The MSE of $\hat{\theta}$ is defined as:

$$\mathsf{MSE}_\theta\big(\hat{\theta}\big) := E\big[(\hat{\theta} - \theta)^2\big].$$

Furthermore, $\mathsf{MSE}_\theta\big(\hat{\theta}\big) = \mathrm{Var}_\theta\big(\hat{\theta}\big) + \big(\mathsf{Bias}_\theta(\hat{\theta})\big)^2$.

MSE II

Example: Variance of MLE

TODO: Finish example 7.3.3, Casella + Berger

MSE III

Example: Sample variance estimator

TODO: Finish example 7.3.4, Casella + Berger.

MSE IV

- Based on the examples above, should $\hat{\sigma^2}$ be used instead of S^2?
- TODO: See text on 332, Casella + Berger.
- Also, TODO: 7.3.5 is kind of a cool example... Include it?

Sufficiency

NAME

Efficiency

NAME

References and Acknowledgements

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